Correlation Analysis

Definition: *An analysis of the covariation of two or more variables is called correlation*.

The problem of analysing the relation between different series can be broken down into three steps:

i) Determining whether a relation exists and, it if does, measuring it.

ii) Testing whether it is significant.

iii) Establishing the cause and effect relation, if any.

The measure of correlation, called the ***correlation coefficient***.

Types of Correlation:

(i) Positive and Negative

(ii) Simple, Partial and Multiple

(iii) Linear and Non-linear

Methods of Studying Correlation:

I. Scatter Diagram

II. Graphical Method

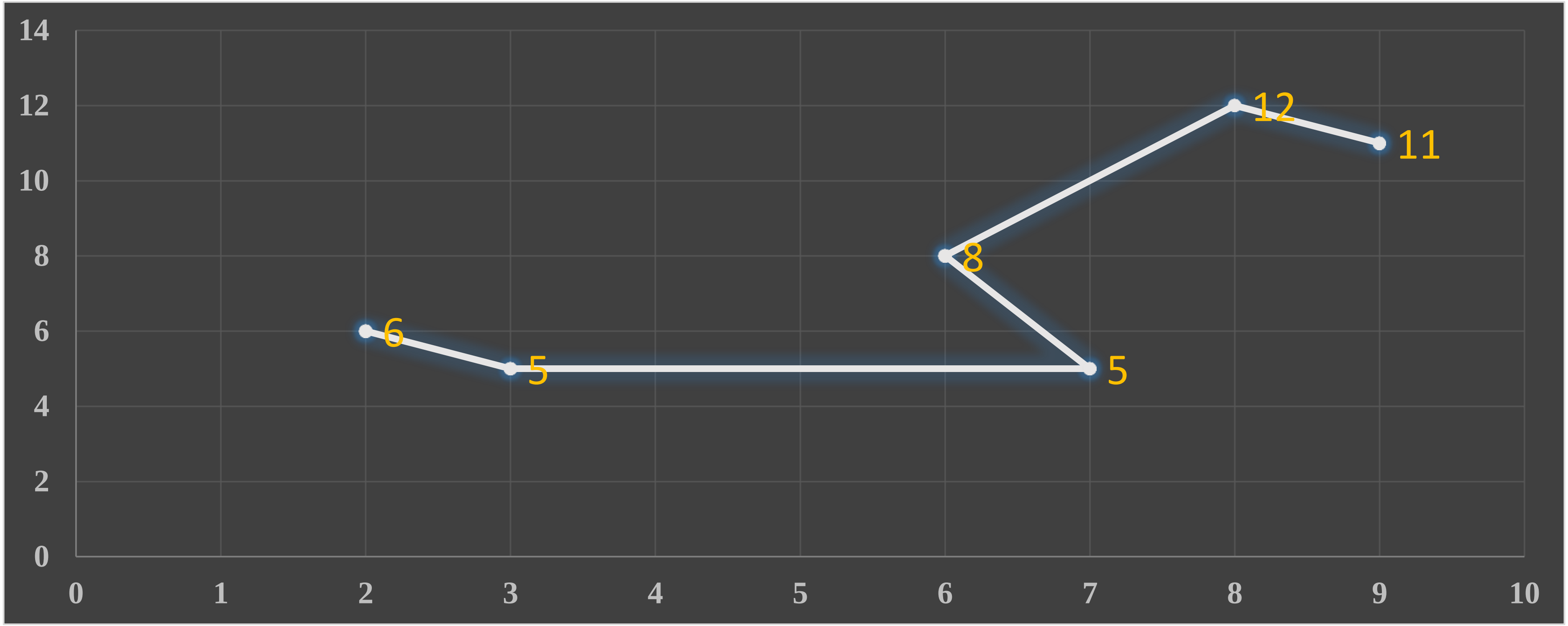
III. Karl Pearson’s Coefficient of Correlation

IV. Spearman’s Rank Correlation

Scatter Diagram

Make a scatter diagram for the following data and find the correlation between the two variables X and Y?

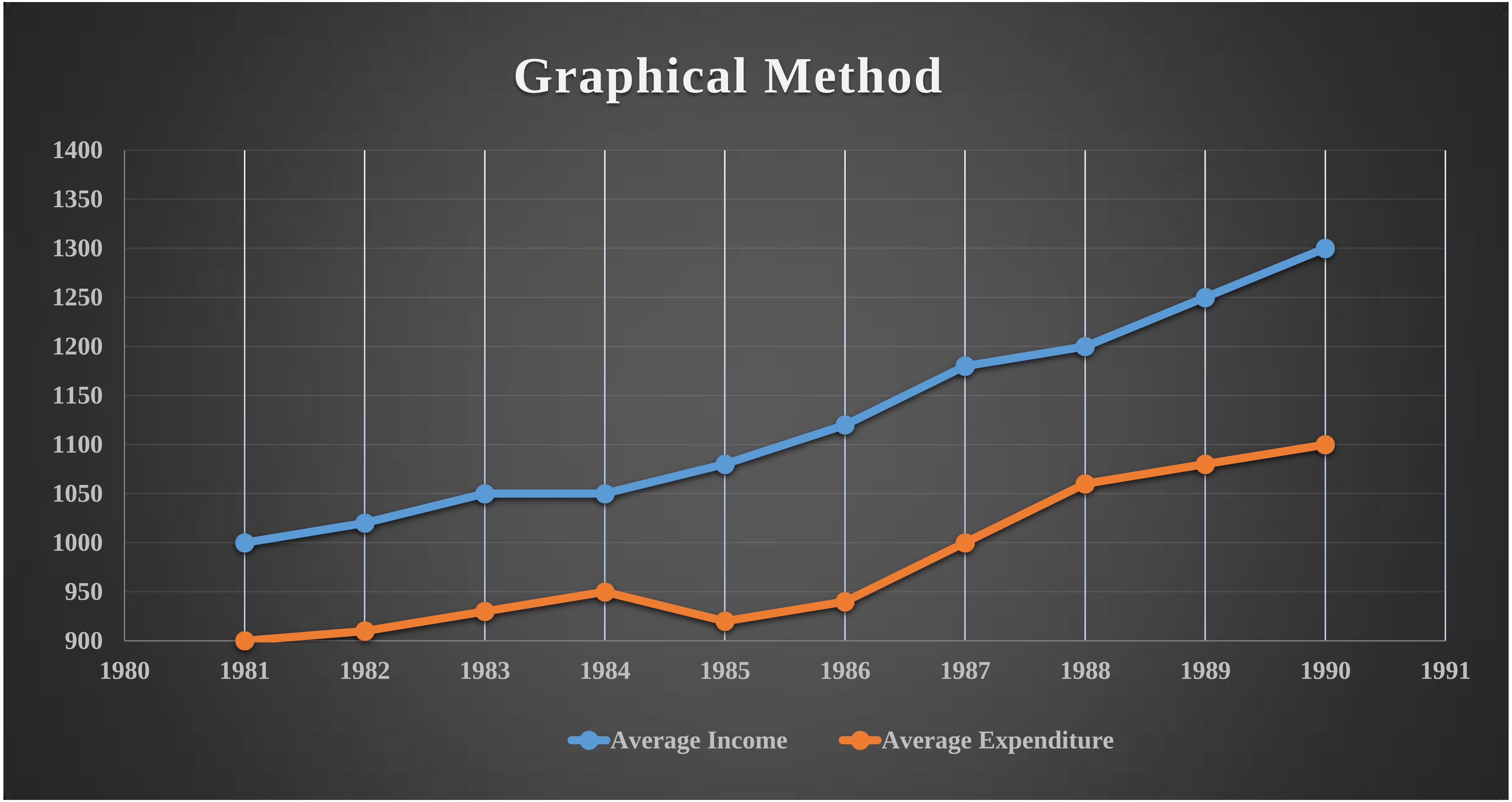
X	2	3	7	6	8	9
Y	6	5	5	8	12	11



Graphical Method

From the following data ascertain whether the income and expenditure of the 100 workers of a factory are correlated:

Year	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
Average Income (in Rs.)	1000	1020	1050	1050	1080	1120	1180	1200	1250	1300
Average Expenditure (in Rs.)	900	910	930	950	920	940	1000	1060	1080	1100


Graphical Method
1400
1350
1300
1250
1200
1150
1100
1050
1000
950
900
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
Average Income
Average Expenditure

Karl Pearson's Coefficient of Correlation

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2}\ \sqrt{\sum(Y - \bar{Y})^2}}$$

This method is to be applied only when the deviations of items are taken from ***actual mean***.

Example 3

Calculate the Karl pearson's coefficient of correlation from the following data:

X	6	8	12	15	18	20	24	28	31
Y	10	12	15	15	18	25	22	26	28

X	***X-18***	$(X-18)^2$	***Y***	***Y-19***	$(Y-19)^2$	**(*X-18*) (*Y-19*)**
6	-12	144	10	-9	81	108
8	-10	100	12	-7	49	70
12	-6	36	15	-4	16	24
15	-3	9	15	-4	16	12
18	0	0	18	-1	1	0
20	2	4	25	6	36	12
24	6	36	22	3	9	18
28	10	100	26	7	49	70
31	13	169	28	9	81	117
162	0	598	171	0	338	431

$$r = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum(X-\bar{X})^2}\sqrt{\sum(Y-\bar{Y})^2}} = \frac{431}{\sqrt{598 \times 338}} = +0.959$$

When deviations are taken from ***assumed mean***:

$$r = \frac{\sum d_x d_y - \frac{(\sum d_x)(\sum d_y)}{N}}{\sqrt{\sum d_x^2 - \frac{(\sum d_x)^2}{N}}\ \sqrt{\sum d_y^2 - \frac{(\sum d_y)^2}{N}}}$$

Where, $d_x = X - A$

$d_y = Y - A$

$\boldsymbol{N}$ = Total number of observations

$\boldsymbol{A}$ - assumed mean

Example 4

Calculate the correlation coefficient for the following data:

X	45	55	56	58	60	65	68	70	75	80	85
Y	56	50	48	60	62	64	65	70	74	82	90

Example 5

Following table gives the distribution of the total population and those who are wholly or partially blind among them. Find out if there is any relation between age and blindness.

Age	**No. of Persons (in thousands)**	**Blind**
0-10	100	55
10-20	60	40
20-30	40	40
30-40	36	40
40-50	24	36
50-60	11	22
60-70	6	18
70-80	3	15

Age	Mid Points X	$d_x = \frac{(X-35)}{10}$	d_x^2	Blind persons per lakh Y	$d_y = (Y-185)$	d_y^2	$d_x d_y$
0-10	5	-3	9	55	-130	16900	390
10-20	15	-2	4	67	-118	13924	236
20-30	25	-1	1	100	-85	7225	85
30-40	35	0	0	111	-74	5476	0
40-50	45	1	1	150	-35	1225	-35
50-60	55	2	4	200	15	225	30
60-70	65	3	9	300	115	13225	345
70-80	75	4	16	500	315	99225	1260
$N = 8$		$\sum d_x = 4$	$\sum d_x^2 = 44$		$\sum d_y = 3$	$\sum d_y^2 = 157425$	$\sum d_x d_y = 2311$

$$r = \frac{\sum d_x d_y - \frac{(\sum d_x)(\sum d_y)}{N}}{\sqrt{\sum d_x^2 - \frac{(\sum d_x)^2}{N}}\sqrt{\sum d_y^2 - \frac{(\sum d_y)^2}{N}}} = 0.898$$

Correlation for grouped data:

$$r = \frac{\sum f d_x d_y - \frac{(\sum f d_x)(\sum f d_y)}{N}}{\sqrt{\sum f d_x^2 - \frac{(\sum f d_x)^2}{N}} \sqrt{\sum f d_y^2 - \frac{(\sum f d_y)^2}{N}}}$$

Example 6:

Calculate the Karl pearson's coefficient of correlation and its probable error between ages of 100 families from the following data:

Ages of Parents in years	**Ages of Children's in years**					
	5-10	10-15	15-20	20-25	25-30	Total
15-25	6	3	-	-	-	9
25-35	3	16	10	-	-	29
35-45	-	10	15	7	-	32
45-55	-	-	7	10	4	21
55-65	-	-	-	4	5	9
Total	9	29	32	21	9	**100**

Let X – denote the age of Children’s and Y –denotes the age of Parent's

Y \ X	m	d_x / d_y	5-10	10-15	15-20	20-25	25-30	f	fd_y	fd_y^2	fd_xd_y
	m		7.5	12.5	17.5	22.5	27.5				
			-2	-1	0	1	2				
15-25	20	-2	6 $\boxed{24}$	3 $\boxed{6}$	- $\boxed{0}$	- $\boxed{0}$	- $\boxed{0}$	9	-18	36	30
25-35	30	-1	3 $\boxed{6}$	16 $\boxed{16}$	10 $\boxed{0}$	- $\boxed{0}$	- $\boxed{0}$	29	-29	29	22
35-45	40	0	- $\boxed{0}$	10 $\boxed{0}$	15 $\boxed{0}$	7 $\boxed{0}$	- $\boxed{0}$	32	0	0	0
45-55	50	1	- $\boxed{0}$	- $\boxed{0}$	7 $\boxed{0}$	10 $\boxed{10}$	4 $\boxed{8}$	21	21	21	18
55-65	60	2	- $\boxed{0}$	- $\boxed{0}$	- $\boxed{0}$	4 $\boxed{8}$	5 $\boxed{20}$	9	18	36	28
Total		f	9	29	32	21	9	$N = 100$	-8	122	98
		fd_x	-18	-29	0	21	18	-8			
		fd_x^2	36	29	0	21	36	122			
		fd_xd_y	30	22	0	18	28	98			

$$r = \frac{\sum f d_x d_y - \frac{(\sum f d_x)(\sum f d_y)}{N}}{\sqrt{\sum f d_x^2 - \frac{(\sum f d_x)^2}{N}}\ \sqrt{\sum f d_y^2 - \frac{(\sum f d_y)^2}{N}}}$$

$\sum f d_x d_y = 98,\ \sum f d_x = -8,\ \sum f d_y = -8,$

$\sum f d_x^2 = 122,\ \sum f d_y^2 = 122,\ N = 100$

$$r = \frac{98 - \frac{(-8)\,(-8)}{100}}{\sqrt{122 - \frac{(-8)^2}{100}}\sqrt{122 - \frac{(-8)^2}{100}}} = +0.802$$

Rank Correlation Coefficient

This measure is useful when quantitative measures for certain factors (such as the evaluation of leadership of the judgement) cannot be fixed, but the individuals in the group can be arranged in order thereby obtaining for each individual a number indicating his (her) rank in the group.

Spearman's rank correlation coefficient is defined as :

$$R = 1 - \frac{6 \sum D^2}{N^3 - N}$$

R – rank correlation coefficient

D – difference of the ranks between paired items in two series.

When Actual Ranks are given:

(i) Take the differences of the two ranks, *i.e.*, $(\boldsymbol{R_1 - R_2})$ and denote these differences by $\boldsymbol{D}$.

(ii) Square these differences and obtain the total.

(iii) Apply the formula: $\boldsymbol{R = 1 - \frac{6 \sum D^2}{N^3 - N}}$

Example 6

Two judges in a beauty competition rank the 12 entries as follows:

X	1	2	3	4	5	6	7	8	9	10	11	12
Y	12	9	6	10	3	5	4	7	8	2	11	1

What degree of agreement is there between the judgement of the two judges ?

Example 7

Ten competitors in a beauty contest are ranked by three judges in the following order:

1st Judge	1	6	5	10	3	2	4	9	7	8
2nd Judge	3	5	8	4	7	10	9	1	6	9
3rd Judge	6	4	9	8	1	6	3	10	5	7

Use the rank correlation coefficient to determine which pairs of judges has the nearest approach to common tastes in beauty.

When Ranks are not given:

Ranks can be assigned by taking either the highest value as 1 or the lowest value as 1.

Example 8:

X	53	98	95	81	75	61	59	55
Y	47	25	32	37	30	40	39	45

X	**R_1**	**Y**	**R_2**	**D^2**
53	1	47	8	49
98	8	25	1	49
95	7	32	3	16
81	6	37	4	4
75	5	30	2	9
61	4	40	6	4
59	3	39	5	4
55	2	45	7	25
				$\sum D^2 = 160$

Equal Ranks

Where equal ranks are assigned to some entries an adjustment in the above formula for calculating the rank coefficient of correlation is made.

$$R = 1 - \frac{6\left\{\sum D^2 + \frac{1}{12}(m^3 - m) + \frac{1}{12}(m^3 - m) + \cdots \ldots\right\}}{N^3 - N}$$

m - number of items whose ranks are common

Example 9

Calculate the coefficient of rank correlation from the following data:

X	48	33	40	9	**16**	**16**	65	24	**16**	57
Y	**13**	**13**	24	**6**	15	4	20	9	**6**	19

X	**R_1**	**Y**	**R_2**	**D^2**
48	8	**13**	5.5	6.25
33	6	**13**	5.5	0.25
40	7	24	10	9.00
9	1	**6**	2.5	2.25
16	3	15	7	16.00
16	3	4	1	4.00
65	10	20	9	1.00
24	5	9	4	1.00
16	3	**6**	2.5	0.25
57	9	19	8	1.00
				$\sum D^2 = 41$

$$R = 1 - \frac{6\left\{\sum D^2 + \frac{1}{12}(m^3 - m) + \frac{1}{12}(m^3 - m) + \cdots\ldots\right\}}{N^3 - N}$$

$$R = 1 - \frac{6\left\{41 + \frac{1}{12}(3^3 - 3) + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(2^3 - 2)\right\}}{10^3 - 10} = 0.733$$

Probable Error:

With the help of the probable error it is possible to determine the reliability of the value of the coefficient in so far as it depends on the conditions of random sampling.

$$P.E. = 0.6745\left(\frac{1-r^2}{\sqrt{N}}\right)$$

Where ***r*** **-** is the coefficient of correlation and ***N*** **-**is the number of pairs of items.

Limitations

1) If the value of $r-$ is less than the probable error there is no evidence of correlation, *i.e.* the value of $r-$ is not at all significant.

2) If the value of $r-$ is more than six times the probable error, the existence of correlation is practically certain, *i.e.*, the value of $r-$ is significant.

3) By adding and subtracting the value of probable error from the coefficient of correlation we get limits for ***population correlation coefficient***.

$$\rho = r \pm P.E.$$

$\rho -$ correlation in the population

$r -$ correlation in the sample from the population

$$P.E. = 0.6745\ \left(\frac{1-r^2}{\sqrt{N}}\right)$$

$$r = +0.802\ ,\ N = 100$$

$$P.E. = 0.6745\ \left(\frac{1-(0.802)^2}{\sqrt{100}}\right) = \ 0.024$$

The limits of the Population Correlation Coefficient are :

$$\rho = 0.802 \pm 0.024$$

$$(0.778, 0.826)$$

Coefficient of Determination (r^2)

$r^2 -$ The square of the coefficient of correlation.

If $r = 0.9$ then $r^2 = 0.81$ and this would mean that **81%** of the variation in the dependent variable has been explained by the independent variable.

Definition: The ratio of the explained variance to the total variance.

$$Coefficient\ of\ Determination\ = \frac{Explained\ Variation}{Total\ Variance}$$

Partial Correlation

Partial Correlation coefficient provides a measure of the relationship between the dependent variable and other variables, with the effect of the most of the variables eliminated.

If we denote by $\boldsymbol{r_{12.3}}$ the coefficient of partial correlation between X_1 and X_2, keeping X_3 constant,

$$\boldsymbol{r_{12.3}} = \frac{\boldsymbol{r_{12} - r_{13}\, r_{23}}}{\sqrt{(\boldsymbol{1 - r_{13}^2})}\sqrt{(\boldsymbol{1 - r_{23}^2})}}$$

Similarly, $\boldsymbol{r_{13.2}} = \frac{\boldsymbol{r_{13} - r_{12}\, r_{23}}}{\sqrt{(\boldsymbol{1 - r_{12}^2})}\sqrt{(\boldsymbol{1 - r_{23}^2})}}$ and $\boldsymbol{r_{23.1}} = \frac{\boldsymbol{r_{23} - r_{13}\, r_{12}}}{\sqrt{(\boldsymbol{1 - r_{12}^2})}\sqrt{(\boldsymbol{1 - r_{13}^2})}}$

Multiple Correlation

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}\, r_{13}\, r_{23}}{1 - r_{23}^2}}$$

$$R_{2.13} = \sqrt{\frac{r_{12}^2 + r_{23}^2 - 2r_{12}\, r_{13}\, r_{23}}{1 - r_{13}^2}}$$

$$R_{3.12} = \sqrt{\frac{r_{13}^2 + r_{23}^2 - 2r_{12}\, r_{13}\, r_{23}}{1 - r_{12}^2}}$$